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(Pages : 4)

Name.....

Reg. No.....

**THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MEC 3C 03—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 3C 03—MATHEMATICAL ECONOMICS
(Multiple Choice Questions for SDE Candidates)

1. When $Y''(t) = 10$, $Y(t)$ will be :

- (A) 1. (B) 0.
(C) $5t^2$. (D) $5t^2 + tc_1 + C$.

2. What is the degree of $\left(\frac{dy}{dx^2}\right)^6$.

- (A) 2. (B) 3.
(C) 4. (D) 6.

3. Given $Q_d = 6 - 2P$ $Q_s = 4 + 4P$ equilibrium $\bar{p}$ will be :

- (A) 6. (B) 4.
(C) 10. (D) $\frac{1}{3}$

4. Law of returns to scale is a theory pertaining to :

- (A) Market period. (B) Short period.
(C) Long period. (D) None of the above.

5. If $\frac{dy}{dt} = 15$, the value of $Y(t)$ is :

- (A) 15. (B) 0.
(C) $15t + A$. (D) None of the above.

6. Which of the following is used for constrained optimization ?

- (A) Hessian. (B) Barclered Hessian.
(C) Discriminant. (D) Jacobian.

7. The value of Y when $\frac{dy}{dt} = Y^2 t$ is :
- (A) $\frac{2}{t+c}$. (B) $\frac{-2}{t+c}$.
- (C) $\frac{1}{t^2+c}$. (D) $\frac{-2}{t^2+c}$.
8. Production function shows technological relationship between input and ____.
- (A) Output (B) Factors of production.
- (C) Both of the above. (D) None of the above.
9. When demand for a good is give by $Q = 40 - P$, the maximum amount that would be demanded at nil price is :
- (A) 1. (B) 0.
- (C) 40. (D) 400.
10. Technical relationship between input and output is called :
- (A) Elasticity. (B) Production function.
- (C) Input function. (D) None of the above.
11. For the production function $Q = AL^{0-4} L^{05}$, which of the following is true ?
- (A) Increasing returns to scale. (B) Diminishing returns to scale
- (C) Constant returns to scale. (D) Variable returns to scale.
12. What is the order of $\left(\frac{d^3 y}{dx^3}\right)^4 + \left(\frac{d^2 y}{dx^2}\right)^6 = 10 - Y$.
- (A) First order. (B) Second order.
- (C) Third Order. (D) Fourth order
13. What is the degree of $\left(\frac{d^3 y}{dx^3}\right)^4 + \left(\frac{d^2 y}{dx^2}\right)^6 = 10 - Y$.
- (A) 3. (B) 4.
- (C) 2. (D) 6.

14. The producer will be in equilibrium when :

- (A) $MRTS = \frac{p_x}{p_y}$. (B) $MRTS > \frac{p_x}{p_y}$.
 (C) $MRTS < \frac{p_x}{p_y}$; (D) None of the above.

15. The longrun theory of output behavior is known as :

- (A) Diminishing returns. (B) Law of variable proportions.
 (C) Diminishing Marginal products. (D) Law of returns to scale.

16. Select the odd one among land, capital, organization, profit :

- (A) Land. (B) Capital.
 (C) Organisation. (D) Profit.

17. In the production function $Q = r \left[\delta C^{-\alpha} + (1-\delta) N^{-\alpha} \right]^{\frac{-V}{\alpha}}$, efficient is measured with :

- (A) δ . (B) r .
 (C) N . (D) C .

18. The value of Rs. 100 at 10 % interest for two years :

- (A) 110. (B) 111.
 (C) 121. (D) 130.

19. Functional relationship between input and output is called :

- (A) Isoquants.. (B) Isocost.
 (C) Input function. (D) Production function.

20. What is the degree of $\frac{d^3Y}{dx^3} + x^2Y \left(\frac{d^2Y}{dx^2} \right) - 4Y^4 = 1$.

- (A) First. (B) Second.
 (C) Third. (D) Fourth.

**THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MEC 3C 03—MATHEMATICAL ECONOMICS

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least **eight** questions.*

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

1. Distinguish between the order and degree of a difference equation with example.
2. What do you mean by CES production function ?
3. Discuss the concept of Expansion Path.
4. What is Economic Region of production ?
5. Discuss the concept of Returns to Scale.
6. Limitations of Cobb-Douglas Production Function.
7. How will you demonstrate Producer's Equilibrium ?
8. What is elasticity of substitution ?
9. Why do you think that investment appraisal is necessary ? What is Internal Rate of Return ?
10. What is the difference between risk and uncertainty in Economics ?
11. What do you mean by sensitivity analysis ?
12. What do you mean by separation of Variables ?

(8 × 3 = 24 marks)

Turn over

Section B

*Answer at least **five** questions.*

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Solve the first order homogenous linear difference equation $y_{x+1} = 4y$.
14. Explain the Probability Distribution approach in evaluation risk.
15. Derive the formula for the value P_t of an initial amount of money P_0 deposited at i interest for t years when compounded annually.
16. Explain the Lagged Income Determination model using difference equations.
17. Briefly describe the economic application of differential equations in Economics.
18. What is the difference between Risk adjusted discount rate (RAD) approach and Certainty-Equivalent approach ?
19. Explain homogeneous production function.

(5 × 5 = 25 marks)

Section C

*Answer any **one** question.*

Each question carries 11 marks.

20. What is production function? Explain the Law of Variable Proportions.
21. Optimise the generalized Cobb-Douglas production function $Q = K^{0.3} L^{0.5}$ subject to $6K + 2L = 384$.

(1 × 11 = 11 marks)